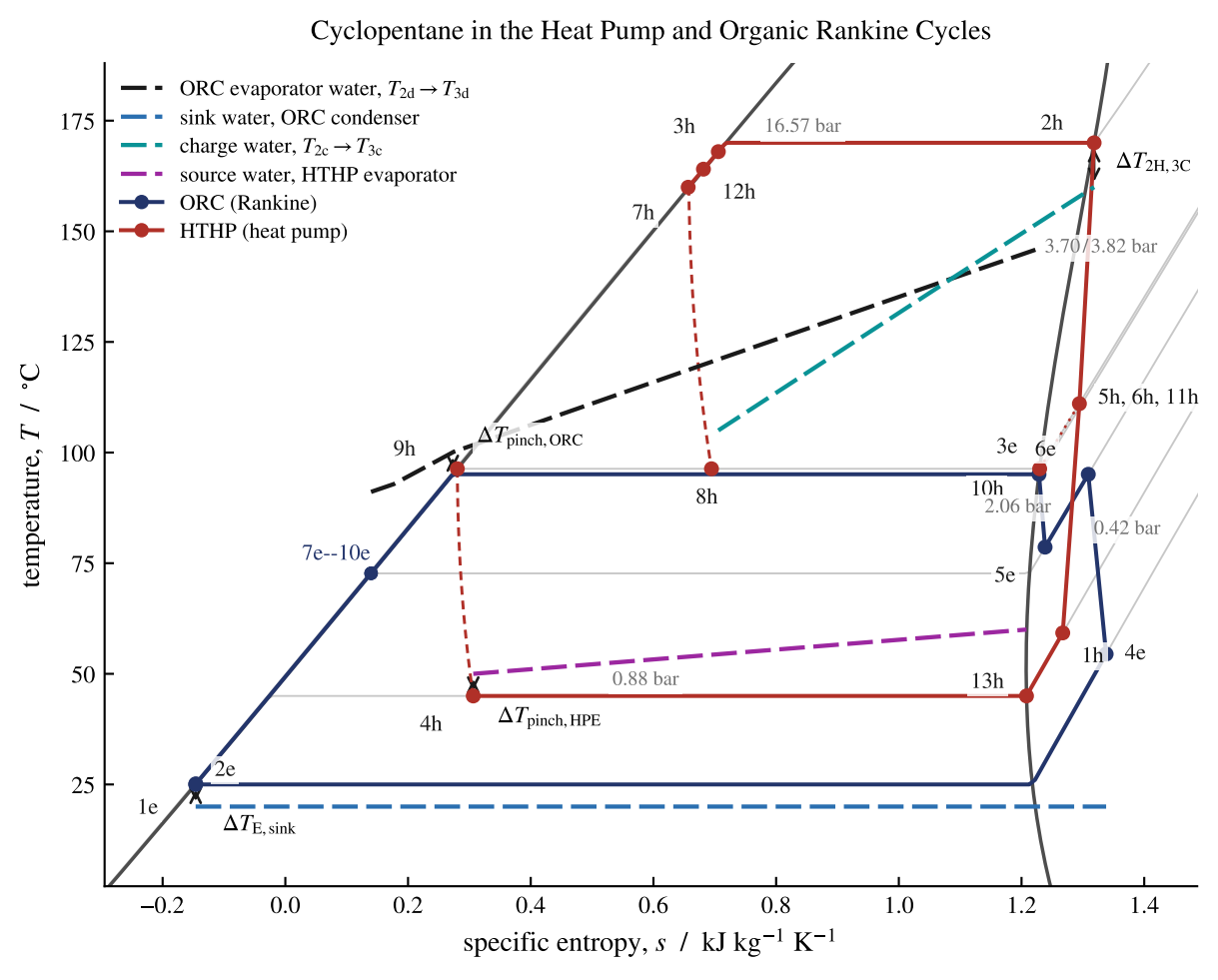
Cyclopentane in the Heat Pump and Organic Rankine Cycles
ORC evaporator water, $T_{2d} \rightarrow T_{3d}$
sink water, ORC condenser
charge water, $T_{2c} \rightarrow T_{3c}$
source water, HTHP evaporator
ORC (Rankine)
HTHP (heat pump)
temperature, $T$ / °C
specific entropy, $s$ / kJ kg$^{-1}$ K$^{-1}$
175
150
125
100
75
50
25
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1e
2e
3h
4h
5h, 6h, 11h
7h
7e--10e
8h
9h
10h
12h
13h
1h
2h
3e
4e
5e
6e
16.57 bar
3.70 / 3.82 bar
2.06 bar
0.42 bar
0.88 bar
$\Delta T_{2H,3C}$
$\Delta T_{pinch,ORC}$
$\Delta T_{pinch,HPE}$
$\Delta T_{E,sink}$